MILITARY SPECIFICATION

ADAPTION KIT, ATOMIC WARHEAD TO ROCKET, TRAINING: M87

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of training kit entitled Adaption Kit, Atomic Warhead to Rocket, Training: XM87 which is used for training on the Honest John warhead section.

SPECIFICATIONS

FEDERAL

PPP-I-60 - Tape, Pressure-sensitive adhesive, Waterproof, for Packaging

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1460 - Soldering of Electrical connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8800692 - Tool, Pre-fire.
8821437 - Shaft Half.
8824934 - Shipping and Storage Container XM480E1 for Adaption Kits and Warhead Sections.
8825113 - Nose Base Assembly.
8830816 - Adaption Kit, Atomic Warhead to Rocket, Training: M87.
8830821 - Nose Base and Adaptor Ring Assembly.
EL-8830816 - Equipment List.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The training kit shall comply with all requirements specified on the drawing (dwg) 8830816, with all requirements specified herein, and with requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Pre-fire. The training kit shall cause the T-4162 test set to give both go and no-go indications when tested as specified in 4.4.1.1.

3.6 Storage. The training kit shall give both go and no-go indications when tested as specified in 4.4.1.2.

3.7 Operational requirements.

3.7.1 Starting torque (test set(see note 1. a. dwg 8821437). The starting torque shall not exceed 20 inch pounds and the average of 5 trials shall not exceed 15 inch pounds when tested as specified in 4.4.2.2.

3.7.2 Setting torque (timer). The torque required to set the timer shall not exceed 15 inch pounds when tested as specified in 4.4.2.2.

3.7.3 Torque (fuze option selector). The switch shall operate with a torque not exceeding 20 inch pounds when tested as specified in 4.4.2.3.

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3.8 Environmental testing. The training kit shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The training kit shall also comply with 4.4.1 and 4.4.2 before and after each environmental test. Testing shall be as specified in 4.4.3. Tests shall be conducted in the order specified.

3.8.1 Vibration. The training kit shall withstand vibration through the frequency range of 5 to 55 Hertz (Hz) at a double amplitude of 0.06 plus or minus (±) 0.006 inch for a period of 30 minutes in each of three major mutually perpendicular axes and after test shall comply with 3.5, 3.6 and 3.7. Testing shall be as specified in 4.4.3.1.

3.9 Paint adhesion. The painted surfaces of the training kit shall show no evidence of peeling, flaking or other deterioration when tested as specified in 4.4.4.

3.10 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460 (MU).

3.11 Workmanship.

3.11.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.11.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.11.3 Burrs. Parts shall be free from burrs.

3.11.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.11.5 Cleaning. The cleaning method shall not be injurious to any of the parts, not shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 8830816) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

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4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of Characteristics (see 4.3.2.1)		X
2. Pre-fire (see 3.5)		X
3. Storage (see 3.6)		X
4. Operational requirements (see 3.7)		X
5. Vibration (see 3.9)	X	
6. Paint adhesion (see 3.9)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

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4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Assembly nose base (see dwg 8825113 covering a detail of dwg 8830816).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101. Incorrect number of inserts	Visual
	102. Any component loose, damaged or missing	Visual

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Minor:

201. Evidence of poor workmanship (see 3.11).....Visual

4.3.2.1.2 Assembly nose base with adaptor ring assembled (see dwg 8830821 covering a detail of dwg 8830816).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Seal loose, damaged or missing	Visual-Manual
102.	Hardware loose, damaged or missing	Visual-Manual
103.	Any component loose, damaged or missing	Visual-Manual
104.	Marking missing or incomplete	Visual-Manual
Minor:		
201.	Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.3 Adaption Kit assembly complete (see dwg 8830816).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Nose bolt loose, damaged or missing	Visual-Manual
102.	Packing insufficient	Visual
103.	Hardware loose, damaged or missing	Visual-Manual
104.	Any operation inadequate or missing	Visual
105.	Any component loose, damaged or missing	Visual-Manual
106.	Soldering in accordance with MIL-STD-1460 (MU)	Visual
107.	Objective evidence relative to material missing or improper (see 3.1)	Visual
108.	Serialization improper or missing (see 3.4)	Visual

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Minor:

201. Evidence of poor workmanship (see 3.11).....Visual

4.3.2.1.4 Preparation for delivery (see dwg 8824934).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101. Preservation, packaging and packing in accordance with dwg 8824934	Visual
	102. Marking in accordance with dwg 8824934	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Pre-fire	Major
b. Storage	Major
c. Operational requirements	Major
d. Vibration	Major
e. Paint adhesion	Major

Of these tests, d is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Pre-fire (see 3.5), Storage (see 3.6), operational requirements (see 3.7) and paint adhesion (see 3.9). Each training kit shall be subjected to these tests (see applicable paragraphs in 4.4). If any item fails to comply with the requirements of 3.5, 3.6, 3.7 and 3.9 the item shall be rejected and removed from the lot.

4.3.3.2 Vibration (see 3.8.1). One training kit shall be selected at random out of every 25 training kits, or portion thereof, produced and subjected to this test (see 4.4.3.1). If the sample fails to comply with the applicable requirement, the lot represented by the sample in which failure occured shall be rejected.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8800630 and EL-8867701). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Continuity.

4.4.1.1 Pre-fire.

4.4.1.1.1 Lamp test. Rotate the prefire test socket counter-clockwise manually at a normal rate using a pre-fire tool (dwg 8800692) both the green lamp (Go) and the red lamp (No-Go) shall light.

4.4.1.1.2 Green lamp test (Go). Rotate the pre-fire test socket clockwise manually at a normal rate using a pre-fire tool (dwg 8800692), and the green lamp (Go) shall light.

4.4.1.2 Storage test. Connect the T4087 Test Set Atomic Weapon to the training adaption kit and perform test in accordance with operating instruction for the T4087 test set.

4.4.2 Operational tests.

4.4.2.1 Starting torque (perform XM). Measure the torque required to turn the generator shaft.

4.4.2.2 Setting torque (timer). The breakaway and maintaining torque necessary to set the unit shall be measured with a torque wrench. In neither case shall the torque exceed 15 inch-pounds. The dial shall not move a measurable distance after setting torque is removed, as determined by visual examination.

4.4.2.3 Torque (fuze option selector). Measure the torque required to operate this switch.

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4.4.3 Environmental tests (see 3.8).

4.4.3.1 Vibration. The training kit shall be mounted to a rigid fixture at station 80.05. The fixture shall be mounted to a suitable vibration-producing apparatus. The training kit shall then be vibrated through a range of frequencies from 5 to 55 Hz at a constant double amplitude of 0.06 + 0.006 inch for a duration of 30 minutes in each of its three (3) major mutually perpendicular axes. Each cycle, 5 to 55 to 5 Hz shall have a duration of one (1) minute and the frequency shall be varied uniformly throughout the cycle. The training kit shall show no damage after the vibration test and shall comply with the requirements of 4.4.1 and 4.4.2.

4.4.4 Paint adhesion. The test shall be performed on all sections of the warhead section, after a minumum of 24 hours drying unless otherwise specified. Water resistant, pressure sensitive adhesive tape (3/4 inch width) conforming to the requirements of PPP-I-60, Type II or III, Grade A or B, shall be used. Press a 2 inch length of a somewhat longer piece firmly on to a flat or cylindrical surface of the item rubbing out all air bubbles under the tape. Allow approximately 10 seconds for the test area to return to room temperature. Grasp a free end of the tape and at a rapid speed strip it from the item or specimen by pulling the tape back upon itself 180 degrees (in such a manner that the tape is folded back to back during the procedure). Observe for bare spots where the paint is removed. Disregard flecks of paint on tape if removal of paint has not been complete through to the metal.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The training kit shall be packaged in accordance with dwg 8824934 using the specified moisture barrier.

5.1.2 Level C. The training kit shall be packaged in accordance with dwg 8824934 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The training kit shall be packed in accordance with dwg 8824934.

5.3 Marking. Marking shall be as specified in accordance with dwg 8824934.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

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6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1783 (Rev 1) Amendment 2 dated 19 February 1965.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A147

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